

---

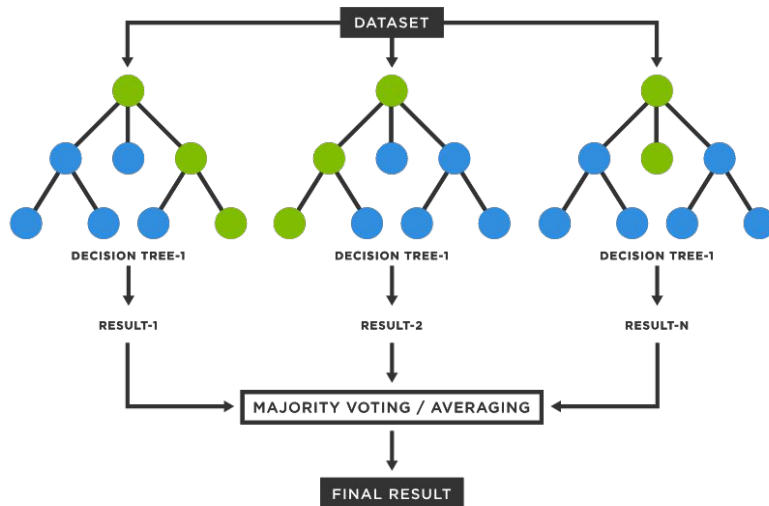
# Meeting #5

July 7, 2026

# Training Speech Classifier on TORGO Dataset

## Model Performance

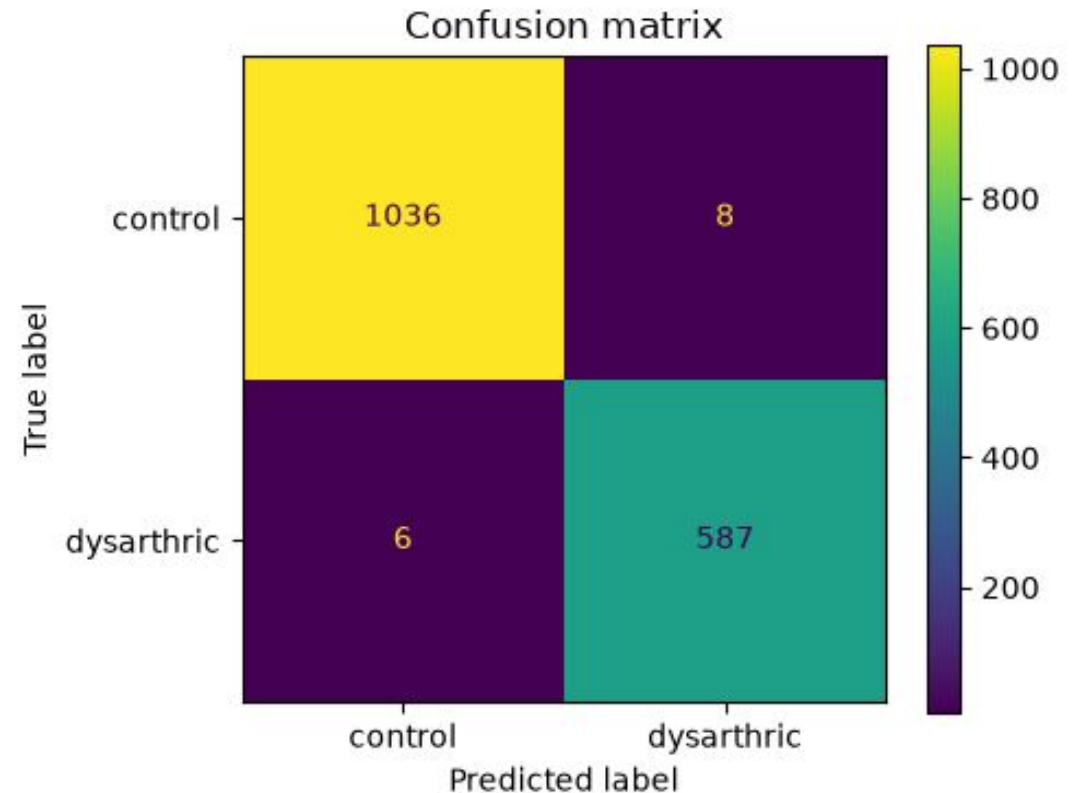
- Random Forest Model from scikit-learn



```
RandomForestClassifier(n_estimators=100,  
class_weight="balanced", random_state=42,  
n_jobs=-1)
```

80/20 Train-Test split

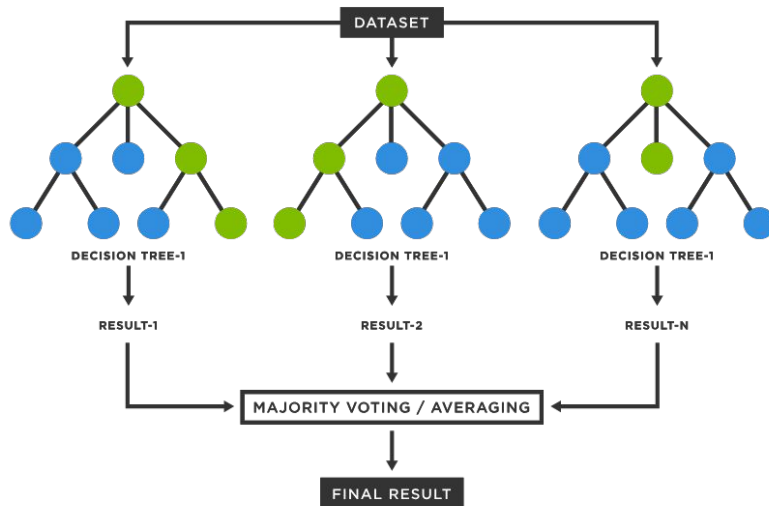
- Before Splitting Train / Test by Speaker
  - Inflated accuracies (99% Accuracy)



# Training Speech Classifier on TORGO Dataset

## Model Performance

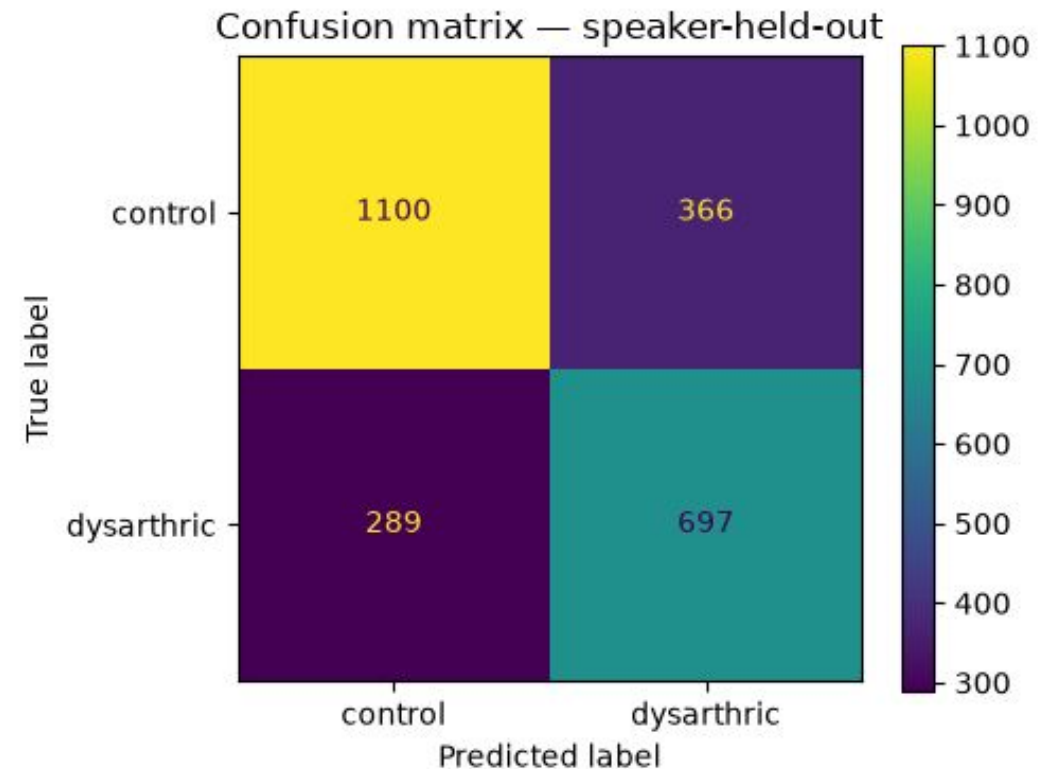
- Random Forest Model from scikit-learn



```
RandomForestClassifier(n_estimators=100,  
class_weight="balanced", random_state=42,  
n_jobs=-1)
```

- 80/20 Train-Test split

- After Splitting Train / Test by Speaker
  - Solves the data leakage problem (73% Accuracy)



# Training Speech Classifier on TORGO Dataset

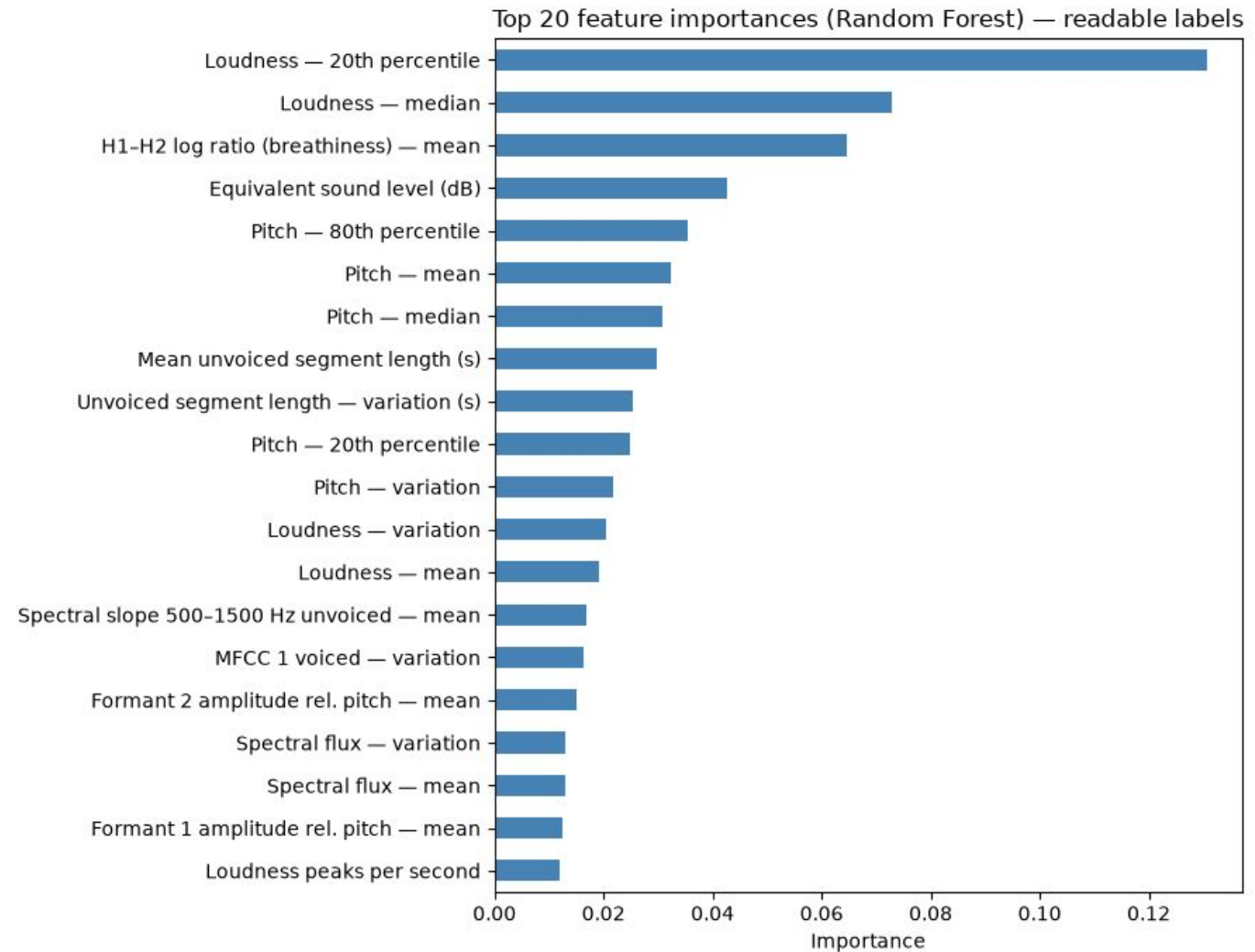
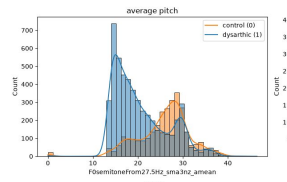
## Feature Importance

- The Random Forest algorithm lets you see which features are the most important for the model's prediction!

*Loudness* - #1, #2, #4

*Breathiness* - #3

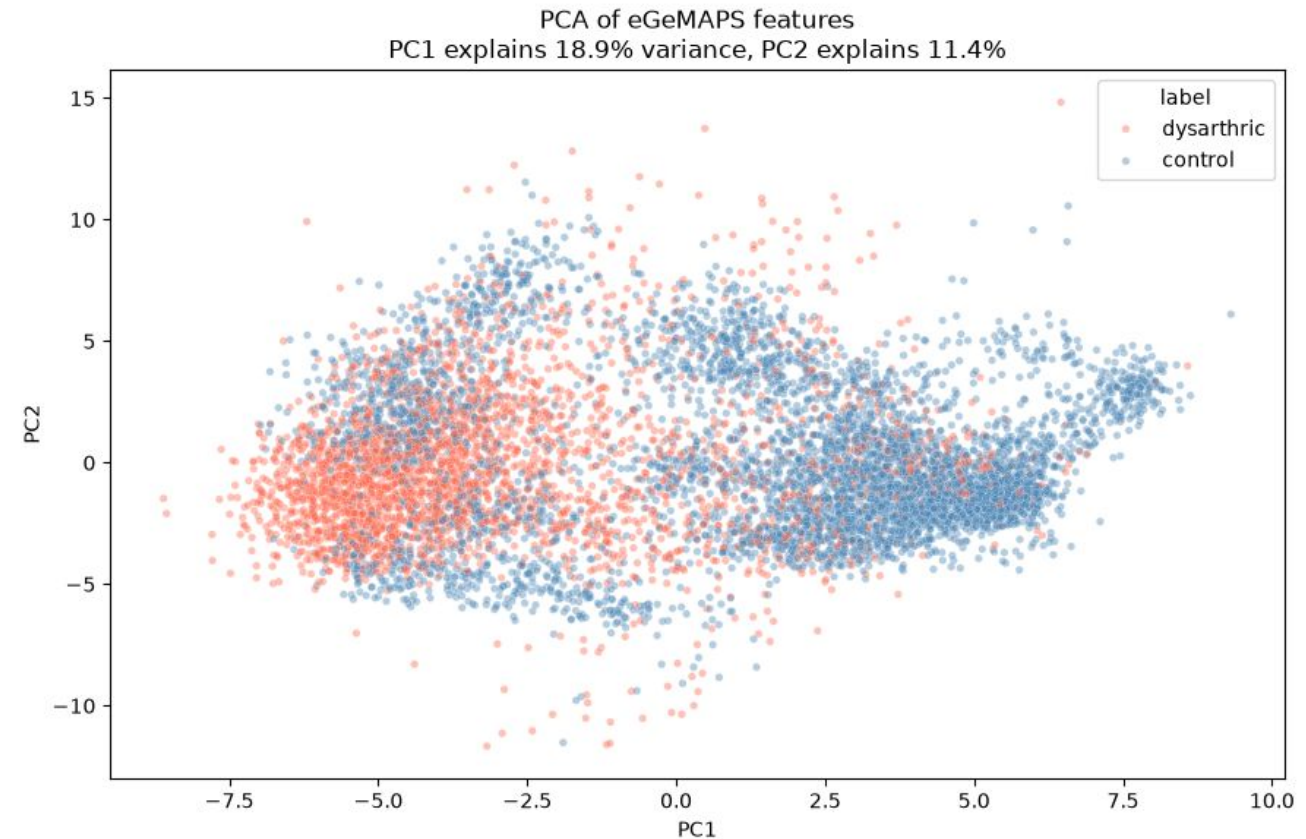
- This gives you important features overall, can use SHAP to look at features important for specific predictions / test cases
- To figure out the sign (direction of effect) of features, can use histogram feature graphs or SHAP beeswarms



# Training Speech Classifier on TORGO Dataset

## PCA

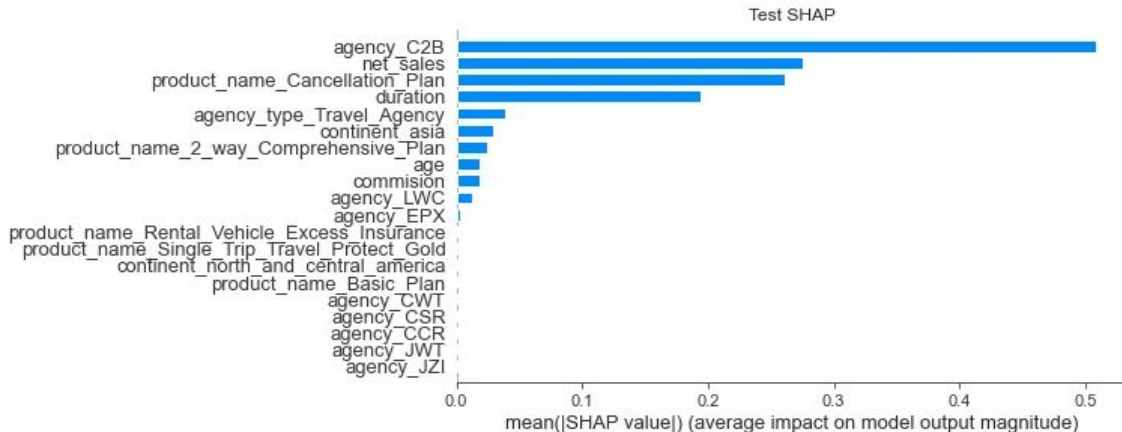
- PCA (Principal Component Analysis) compresses all 88 eGeMAPS features down into 2 numbers per file, called PC1 and PC2.
- These two numbers capture the directions of maximum variance across the whole dataset.
- Optimally, I would expect their points to form two separate clusters
  - Red: dysarthric
  - Blue: control group



# SHAP vs LIME

## SHAP

- SHapley Additive exPlanations
- Uses Shapley values, which measure the contributions of each feature for a particular prediction → can be aggregated globally



## LIME

- Local Interpretable Model-Agnostic Explanations
- Explains the individual predictions of black box by perturbing input data → trains interpretable model on perturbed samples to approximate the black box locally

### Text with highlighted words

Gildan Men's Crew T-Shirts are very comfortable very well made and I would highly recommend them I wear a lot of white shirts and comfort bility and the style and fabric of the shirt is very important to me I Used to always Hans but they've got to be very expensive and I don't mind the price but the quality has also been reduced but Gildan Men's Crew T-Shirts Suppress my expectations and if I could give them a 10 star I would I would highly recommend

# Datasets Access

- **Bridge2AI - *In Progress***
- **Parkinson's Disease - *Downloaded***

## Saarbrücken Voice Database (SVD)

- 2008
- German
- Broadly studies vocal disorders
- Only has 1 subject with "Morbus Parkinson" diagnostic label
- 687 recordings available of "healthy" speakers

## NeuroVoz: a Castilian Spanish corpus

- 2024
- Spanish
- Specific to PD patients who are Castilian Spanish speakers
- 54 PD, 58 HC
- Scripted + spontaneous speech
- 3,000 audio files

## Mobile Device Voice Recordings at King's College London (MDVR-KCL)

- 2017
- English
- Severity annotations in addition to diagnosis
- 16 PD, 21 HC
- Read Text + Spontaneous Dialogue
- 73 audio files

